PROJECT REPORT

Academic Resource Optimization Dashboard

Data-Driven Strategy for Budget Allocation & Student Performance

Week – 2

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1. Executive Summary: Week 2 Focus

This week shifted focus from visual dashboard construction to **Data Integrity and Business Intelligence**. As a Business Analyst, the objective was to ensure the dataset is reliable ("Garbage In, Garbage Out") and to establish the underlying logic that will drive all future institutional decisions.

2. Step-by-Step Data Cleaning Tasks

Before performing any analysis, the following cleaning steps were executed to ensure the data was structured correctly:

- **Standardization:** Subject names were unified to ensure Pivot Tables count "Math" and "math" as the same entity.
- **Logical Column Creation:** * **Status Column:** Added a binary Pass/Fail flag based on a 40-mark threshold.
    - **Grade Column:** Implemented a nested IFS formula to categorize students into A, B, C, D, and F.
    - **Attendance Category:** Segmented students into "High," "Medium," and "Low" attendance to prepare for correlation studies.
- **Validation:** Verified all 50 student records for missing marks or duplicate entries.

3. Business Question Analysis (The 8 Key Insights)

Using the cleaned data, we answered the core questions required by stakeholders:

1. **Highest Class Average: English** remains the strongest subject with a **73.3** average.
2. **Lowest Pass Percentage: Math** is the most critical area, with only a **70.5%** pass rate compared to 88%+ in other subjects.
3. **Overall Topper: STU-030** remains the top performer with **98 marks**.
4. **Attendance Impact:** Analysis confirms a **High Positive Correlation (0.85)**. Students with >85% attendance consistently score in the A-B range.
5. **Overall Success Rate:** The current overall pass rate for the institution is **82%** (9 failures out of 50).
6. **Grade Distribution:** The majority of students fall into the **B and D** ranges, indicating a "split" performance where students are either doing very well or just barely passing.
7. **Performance Anomalies:** No students with "High Attendance" failed, confirming that attendance is the single best predictor of success.
8. **Subject Difficulty: Math** is identified as the most challenging subject based on the lowest average score (**55.1**).

4. Final Business KPI List

These metrics are now calculated and ready to be visualized in Week 3:

- **Class Average Score:** 64.38% (A baseline for teaching effectiveness).
- **Subject Difficulty Index:** Math (High Difficulty) vs. English (Low Difficulty).
- **Attendance vs. Marks Correlation:** 0.85 (Proves behavioral impact).

- **Low-Performing Student %:** 18% of the student body requires urgent intervention.
- **High Distinction %:** 16% of students achieved an "A" grade.

5. Strategic Business Insight

The data shows that **resource scarcity is not the problem—resource allocation is.** * **The Problem:** The institution is currently treating all subjects equally.

- **The Evidence:** Math accounts for over 50% of all failing grades.
- **The Solution:** By correlating attendance and marks, we have identified that an early intervention program triggered at **75% attendance** could prevent 80% of current failures before they happen.

Week 2 Conclusion

The data is now **clean, structured, and logically sound**. We have moved beyond "just looking at marks" to understanding the **behavioral causes** (Attendance and Prep Courses) behind the grades. This foundation ensures that the Week 3 Dashboard will provide meaningful, accurate recommendations for budget reallocation.